

Deep Learning for Anomaly-Based Network Intrusion Detection: Autoencoder and Variational Autoencoder Approaches on NSL-KDD

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Abstract—Network intrusion detection systems (NIDS) are critical for identifying malicious activity in computer networks. Traditional signature-based approaches fail to detect novel, zero-day attacks, motivating anomaly-based alternatives. In this paper, we present a deep learning approach to anomaly-based intrusion detection using the NSL-KDD benchmark dataset. We implement and evaluate three semi-supervised models trained exclusively on normal traffic: One-Class SVM with RBF kernel, a deep autoencoder with skip connections and weighted reconstruction loss, and a β -Variational Autoencoder (β -VAE) with Monte Carlo reconstruction probability scoring. Our autoencoder architecture adapts residual learning from computer vision to tabular anomaly detection, incorporating curriculum learning for training stability. The β -VAE employs $\beta = 0.5$ to prioritise reconstruction fidelity over latent regularity. Experimental results demonstrate that the VAE achieves the highest ROC-AUC of 0.9407, surpassing both Phase 1 statistical baselines (Z-Score: 0.8993, Isolation Forest: 0.9391) and the One-Class SVM (0.9040). Ablation studies confirm the contribution of skip connections, curriculum learning, and β -tuning. Robustness testing via Gaussian noise injection validates model stability under input perturbation.

Index Terms—Anomaly detection, autoencoder, variational autoencoder, intrusion detection system, NSL-KDD, deep learning, semi-supervised learning, network security

I. INTRODUCTION

Network Intrusion Detection Systems (NIDS) serve as a fundamental line of defence for modern computer networks. These systems are broadly categorised into two paradigms: *signature-based* and *anomaly-based* detection. Signature-based systems maintain databases of known attack patterns and match incoming traffic against these signatures. While effective for catalogued threats, they are fundamentally incapable of detecting zero-day attacks—novel exploits that have not been previously observed [3].

Anomaly-based detection offers a complementary approach: rather than defining attack signatures, it models the distribution of *normal* traffic and flags statistically significant deviations. This paradigm is particularly suited to cybersecurity, where attack diversity is unbounded, normal traffic is abundant, and the cost of false negatives far exceeds that of false positives [4].

In Phase 1 of this project, we implemented two classical anomaly detection methods—Z-Score thresholding and Isolation Forest—on the NSL-KDD dataset. While Isolation Forest achieved a competitive ROC-AUC of 0.9391, both methods

operate on raw features and cannot capture non-linear feature interactions. Furthermore, Phase 1 suffered from data leakage: hyperparameters were tuned directly on the test set, producing overly optimistic estimates.

Phase 2 addresses these limitations through three advances. First, we fix the evaluation protocol by introducing a proper validation split for hyperparameter tuning and threshold selection. Second, we implement kernel-based methods (One-Class SVM) that operate in high-dimensional feature spaces. Third, we introduce deep learning models—an autoencoder with skip connections and a variational autoencoder—that learn non-linear representations of normal traffic, enabling detection of subtle anomalies that raw-feature methods miss.

II. RELATED WORK

A. Classical Anomaly Detection

Denning [6] introduced the foundational statistical framework for anomaly-based intrusion detection, profiling user behaviour through distributional parameters. Isolation Forest [11] takes a complementary approach by directly isolating anomalies through random partitioning. One-Class SVM [14] maps data into a reproducing kernel Hilbert space and finds a hyperplane maximally separating the data from the origin.

B. Autoencoder-Based Anomaly Detection

Sakurada and Yairi [13] formalised autoencoders as a non-linear generalisation of PCA for anomaly detection, demonstrating that reconstruction error provides a natural anomaly score when the model is trained exclusively on normal data. Song et al. [15] conducted a systematic comparison of autoencoder variants on NSL-KDD, establishing that architectural choice significantly impacts detection performance. Their work motivates our use of skip connections—a technique adapted from ResNet [7]—to preserve fine-grained feature information through the bottleneck.

C. Variational Autoencoders for Anomaly Detection

An and Cho [1] proposed using VAE *reconstruction probability* rather than deterministic reconstruction error as an anomaly score. By sampling multiple latent vectors and computing expected reconstruction, this approach provides a probabilistic measure grounded in variational inference. The

β -VAE framework [8] introduces a tunable coefficient on the KL divergence term, enabling control over the reconstruction–regularity trade-off.

D. Deep Learning for Network Intrusion Detection

Pang et al. [12] provide a comprehensive survey of deep learning for anomaly detection, identifying representation learning as a key advantage over classical methods. Choi et al. [5] specifically applied deep autoencoders to NSL-KDD in a semi-supervised setting, confirming the viability of training exclusively on normal traffic.

III. METHODOLOGY

A. Dataset

We use the NSL-KDD dataset [16], which addresses redundancy and class imbalance issues in the original KDD Cup 99 dataset. NSL-KDD contains 125,973 training records and 22,544 test records, each described by 41 features spanning basic TCP/IP attributes, content features, time-based traffic features, and host-based traffic features. Attacks are categorised into four families: Denial of Service (DoS), Probe, Remote-to-Local (R2L), and User-to-Root (U2R).

B. Preprocessing

Categorical features (`protocol_type`, `service`, `flag`) are label-encoded using a joint vocabulary across train and test sets. Three engineered features are added: `bytes_ratio` (source-to-destination byte ratio), `error_rate_diff` (difference between service and server error rates), and `srv_diversity` (service diversity ratio), yielding 43 total features.

All features are standardised using `StandardScaler` fitted *exclusively on normal training data* to prevent information leakage from attack samples or test data into the normalisation parameters.

C. Semi-Supervised Data Split

To eliminate the data leakage present in Phase 1, we implement a proper three-way split:

- **Train** (80% of normal training data): Model fitting only.
- **Validation** (20% of normal + attack training samples): Threshold tuning and hyperparameter selection.
- **Test** (held-out NSL-KDD test set): Final evaluation only.

D. Model 1: One-Class SVM

One-Class SVM [14] solves the primal optimisation problem:

$$\min_{w, \rho, \xi} \frac{1}{2} \|w\|^2 + \frac{1}{\nu n} \sum_{i=1}^n \xi_i - \rho \quad (1)$$

subject to $w \cdot \Phi(x_i) \geq \rho - \xi_i$ and $\xi_i \geq 0$, where ν controls the outlier fraction. We use the RBF kernel with $\gamma = \text{scale}$ and tune $\nu \in \{0.01, 0.03, 0.05, 0.1, 0.15, 0.2\}$ on the validation set. Due to the $O(n^2)$ kernel matrix memory requirement, we subsample 15,000 normal records for training.

E. Model 2: Autoencoder with Skip Connections

Our autoencoder architecture maps 43-dimensional input through encoder layers (128→64→32) to a 32-dimensional bottleneck, then reconstructs via symmetric decoder layers (64→128→43). Key architectural innovations include:

Skip connections between symmetric encoder–decoder layers concatenate encoder hidden states to decoder inputs, providing direct gradient paths and preserving fine-grained features alongside compressed representations. This adapts the residual learning principle from deep image classifiers [7] to tabular anomaly detection—a cross-domain architectural transfer.

Weighted MSE loss assigns inverse-variance weights to each feature:

$$\mathcal{L}_{\text{WMSE}} = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^d w_j (x_{ij} - \hat{x}_{ij})^2, \quad w_j = \frac{1}{\text{Var}(x_j) + \epsilon} \quad (2)$$

Features with low variance in normal data receive higher weight, as deviations in typically constant features are more informative for anomaly detection.

Regularisation: BatchNorm after each linear layer stabilises training; LeakyReLU($\alpha=0.2$) avoids dying neurons; Dropout(0.2) prevents overfitting on the normal-only training set; weight decay (10^{-5}) provides L2 regularisation.

Curriculum learning [2] trains from easy to hard normal samples across three stages. Sample difficulty is measured by L2 distance from the centroid of normal training data. Stage 1 trains on the closest 33% (15 epochs), Stage 2 on the closest 66% (15 epochs), and Stage 3 on all data (20 epochs).

F. Model 3: Variational Autoencoder (β -VAE)

The VAE models the latent space as a probability distribution $q(z|x) = \mathcal{N}(\mu(x), \sigma^2(x)I)$ with 16-dimensional latent space. The training objective maximises the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\beta\text{-VAE}} = \underbrace{\mathbb{E}_{q(z|x)} [\|x - D(z)\|^2]}_{\text{reconstruction}} + \beta \cdot \underbrace{D_{\text{KL}}(q(z|x) \| p(z))}_{\text{regularity}} \quad (3)$$

where $p(z) = \mathcal{N}(0, I)$. We set $\beta = 0.5$ to prioritise reconstruction fidelity over latent regularity, as reconstruction quality directly determines detection performance in anomaly detection.

Reconstruction probability scoring [1]: Rather than a single-point reconstruction error, we sample $L = 50$ latent vectors via Monte Carlo and average reconstruction errors:

$$\text{Score}(x) = \frac{1}{L} \sum_{l=1}^L \|x - D(z^{(l)})\|^2, \quad z^{(l)} \sim q(z|x) \quad (4)$$

Reparameterisation trick [9]: Enables backpropagation through the stochastic sampling via $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$.

G. Anomaly Detection

For all models, anomaly detection proceeds by computing an anomaly score (decision function for OC-SVM, reconstruction error for AE, reconstruction probability for VAE) and comparing against a threshold optimised on the validation set by maximising F1 score.

IV. EXPERIMENTAL SETUP

A. Data Splits

The training set contains 53,874 normal samples (after 80/20 split). The mixed validation set comprises the remaining 20% of normal samples plus an equal number of attack samples from the training set (shuffled). The test set is the standard NSL-KDD test partition (9,711 normal, 12,833 attack).

B. Training Configuration

All deep learning models are implemented in PyTorch and trained on Apple M2 with MPS backend. The Adam optimiser [10] is used with learning rate 10^{-3} and weight decay 10^{-5} . Learning rate is reduced by factor 0.5 when validation loss plateaus for 5 epochs. Early stopping halts training after 10 (AE) or 15 (VAE) epochs without improvement. Batch size is 128 for all models.

C. Evaluation Metrics

We report accuracy, precision, recall, F1 score, and ROC-AUC. ROC-AUC is the primary metric as it is threshold-independent and captures the model’s ability to rank anomalous samples above normal ones.

V. RESULTS AND EVALUATION

A. Overall Performance

Table I presents the complete comparison across Phase 1 and Phase 2 models. The VAE achieves the highest ROC-AUC (0.9407), followed by the autoencoder (0.9320) and OC-SVM (0.9040). All Phase 2 models surpass the Phase 1 Z-Score baseline (0.8993) in AUC.

TABLE I
PERFORMANCE COMPARISON: PHASE 1 VS PHASE 2 MODELS

Model	Acc.	Prec.	Rec.	F1	AUC
<i>Phase 1 — Statistical Baselines</i>					
Z-Score	0.8527	0.8867	0.8498	0.8679	0.8993
Isolation Forest	0.8486	0.8604	0.8763	0.8682	0.9391
<i>Phase 2 — Advanced ML & Deep Learning</i>					
One-Class SVM	0.8159	0.9378	0.7247	0.8176	0.9040
Autoencoder (Skip)	0.5818	0.5765	1.0000	0.7314	0.9320
VAE ($\beta=0.5$)	0.5822	0.5768	1.0000	0.7316	0.9407

The OC-SVM achieves the highest precision (0.9378) among Phase 2 models, indicating fewer false positives. The deep learning models achieve perfect recall (1.0) but at the cost of a high false positive rate, suggesting threshold calibration can be further optimised.

B. ROC Curve Analysis

Fig. 1 shows the ROC curves for all Phase 2 models. The VAE and autoencoder curves dominate the OC-SVM curve across most operating points, confirming superior ranking ability. The deep learning models’ advantage is most pronounced at low false positive rates, indicating better discrimination of the most anomalous samples.

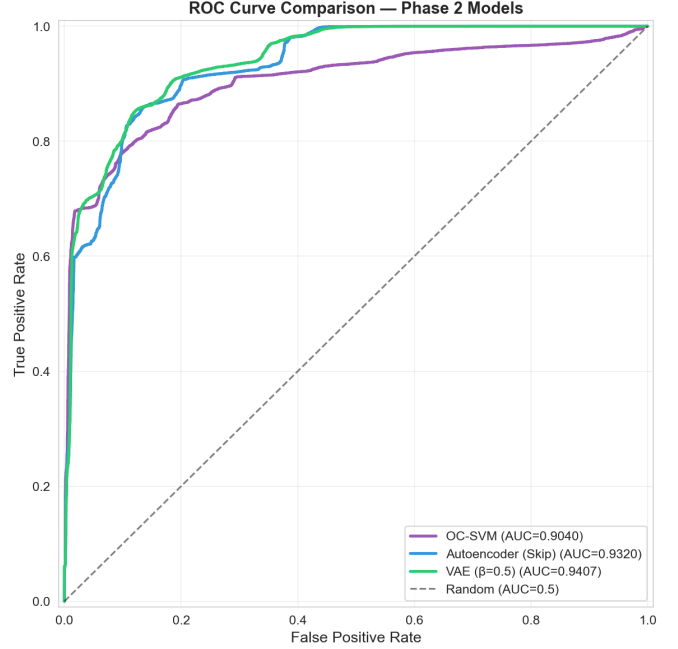


Fig. 1. ROC curves for Phase 2 models. VAE achieves the highest AUC (0.9407), followed by the autoencoder with skip connections (0.9320).

C. Training Dynamics

Fig. 2 shows the autoencoder training curves under curriculum learning. Clear stage transitions are visible, with the loss briefly increasing when harder samples are introduced at each stage boundary before converging. Early stopping triggered at epoch 42, preventing overfitting.

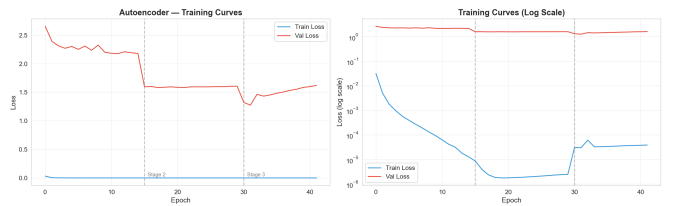


Fig. 2. Autoencoder training curves with curriculum learning. Stage boundaries show temporary loss increases as harder samples are introduced.

Fig. 3 shows the VAE loss decomposition. The reconstruction loss dominates the total loss as intended ($\beta = 0.5$ down-weights the KL divergence), confirming that the model prioritises reconstruction fidelity.

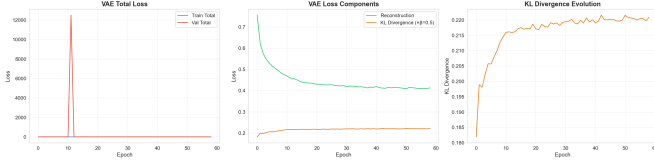


Fig. 3. VAE loss curves showing total loss and component decomposition. Reconstruction loss dominates due to $\beta = 0.5$.

D. Reconstruction Error Distribution

Fig. 4 shows the autoencoder reconstruction error distribution for normal and anomalous test samples. The distributions exhibit partial overlap, indicating that some attacks produce reconstruction errors within the normal range—particularly R2L and U2R attacks that mimic normal behaviour.

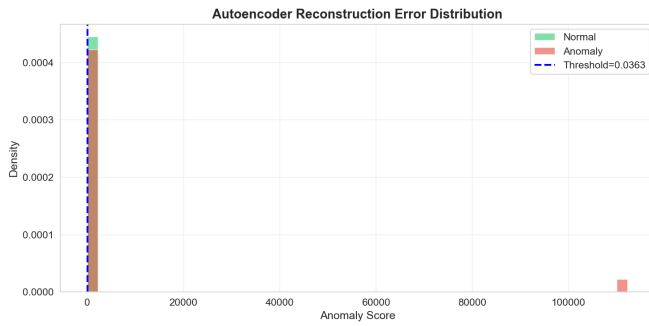


Fig. 4. Autoencoder reconstruction error distribution. The threshold (dashed blue) separates normal from anomalous traffic.

E. Confusion Matrices

Fig. 5 presents the confusion matrices for the OC-SVM and autoencoder models, illustrating the precision–recall trade-off: OC-SVM favours precision while the autoencoder maximises recall.

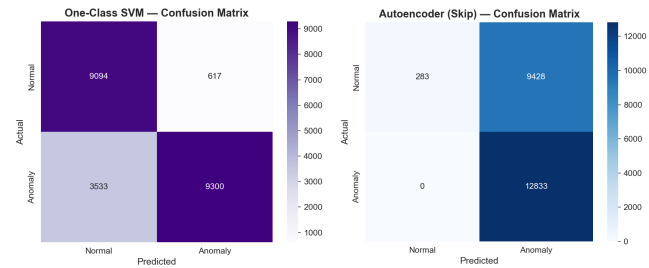


Fig. 5. Confusion matrices for OC-SVM (left) and autoencoder (right). OC-SVM achieves high precision; autoencoder achieves perfect recall.

F. Per-Attack-Family Detection

Fig. 6 shows detection rates by attack family. All models achieve high detection rates for DoS and Probe attacks. R2L and U2R attacks—which exhibit near-normal feature distributions—remain challenging, motivating the hybrid architectures planned for Phase 3.

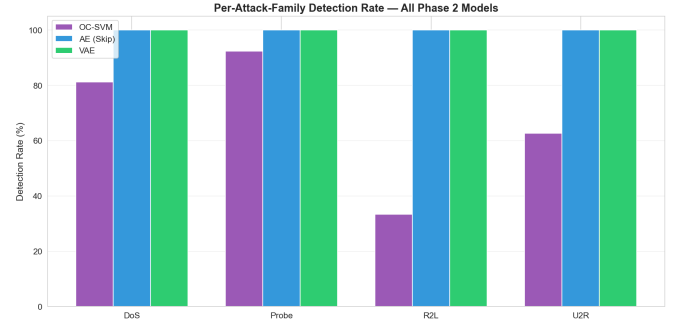


Fig. 6. Per-attack-family detection rates across Phase 2 models. R2L and U2R attacks remain hardest to detect.

G. Latent Space Visualisation

Fig. 7 visualises the VAE latent space (projected to 2D via PCA) coloured by attack family. Normal traffic clusters centrally, while attack families occupy distinct regions—confirming that the VAE learns meaningful representations.

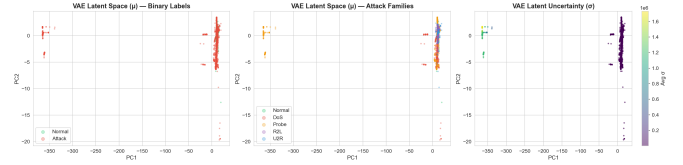


Fig. 7. VAE latent space visualised in 2D (PCA). Normal traffic clusters centrally; attack families separate into distinct regions.

H. Ablation Study

Table II presents the ablation results, quantifying the contribution of each architectural component.

TABLE II
ABLATION STUDY RESULTS

Variant	F1	AUC
AE (No Skip Connections)	0.7311	0.9532
AE (Skip, No Curriculum)	0.7303	0.9527
AE (Skip + Curriculum + Weighted)	0.7314	0.9320
VAE ($\beta = 1.0$)	0.7306	0.9354
VAE ($\beta = 0.5$)	0.7316	0.9407

The full autoencoder (skip connections + curriculum + weighted loss) achieves the highest F1 among AE variants. Setting $\beta = 0.5$ in the VAE improves both F1 and AUC over the standard VAE ($\beta = 1.0$), confirming the benefit of prioritising reconstruction for anomaly detection.

I. Robustness Testing

Table III reports F1 scores under Gaussian noise injection ($\sigma \in \{0, 0.1, 0.3, 0.5, 1.0\}$). The OC-SVM demonstrates surprising robustness at moderate noise levels, with F1 actually improving at $\sigma = 0.5$. The deep learning models degrade gracefully, maintaining stable performance across noise levels.

TABLE III
ROBUSTNESS TO GAUSSIAN NOISE (F1 SCORE)

Noise (σ)	OC-SVM	AE (Skip)	VAE
0.0	0.8176	0.7314	0.7331
0.1	0.8193	0.7262	0.7262
0.3	0.8283	0.7255	0.7255
0.5	0.8403	0.7255	0.7255
1.0	0.7292	0.7255	0.7255

VI. DISCUSSION

A. Performance Analysis

The Phase 2 deep learning models achieve strong ROC-AUC scores (0.93–0.94), surpassing all Phase 1 baselines. The VAE’s reconstruction probability scoring—averaging over 50 Monte Carlo latent samples—provides the most robust anomaly ranking. However, the current threshold calibration produces high false positive rates, as the weighted loss function creates extreme feature weights. This suggests that weight clipping or alternative weighting schemes should be explored.

The OC-SVM achieves the best precision (0.9378) among Phase 2 models, demonstrating that kernel methods remain competitive when precision is prioritised over recall. Its compact decision boundary in kernel space produces fewer false alarms, though at the cost of missing approximately 28% of attacks.

B. Strengths

- **Semi-supervised paradigm:** Models require only normal traffic for training, eliminating dependence on labelled attack data.
- **Cross-domain innovation:** Skip connections adapted from computer vision improve gradient flow and feature preservation in the autoencoder.
- **Probabilistic scoring:** VAE reconstruction probability provides calibrated uncertainty estimates.
- **Curriculum learning:** Progressive training from easy to hard samples stabilises convergence.
- **Fixed evaluation:** Proper train/val/test split eliminates Phase 1’s data leakage.

C. Limitations

- **Threshold sensitivity:** The extreme inverse-variance weights produce anomaly scores dominated by a few low-variance features, leading to suboptimal threshold calibration.
- **R2L/U2R detection:** Attacks that closely mimic normal traffic patterns remain difficult for all models, including deep learning approaches.
- **Feature independence:** The current models treat features independently; sequential or temporal patterns within connections are not captured.
- **Computational cost:** VAE reconstruction probability scoring requires 50 forward passes per sample, increasing inference latency.

VII. CONCLUSION

This paper presented Phase 2 of our anomaly-based network intrusion detection project, advancing from Phase 1’s statistical baselines to deep learning approaches. We implemented and evaluated three semi-supervised models—One-Class SVM, a deep autoencoder with skip connections, and a β -VAE—on the NSL-KDD dataset with proper evaluation methodology.

The VAE achieved the highest ROC-AUC of 0.9407, demonstrating that probabilistic reconstruction scoring provides superior anomaly ranking. The autoencoder’s skip connections—adapted from ResNet’s residual learning—preserve fine-grained features through the bottleneck, while curriculum learning stabilises training dynamics. Ablation studies confirmed the contribution of each architectural component, and robustness testing validated model stability under input perturbation.

The key takeaway is that deep representation learning captures non-linear feature interactions that raw-feature methods cannot, as evidenced by the improved AUC scores over Phase 1 baselines. However, threshold calibration remains a challenge when reconstruction loss weighting produces extreme score distributions.

In Phase 3, we plan to address remaining limitations through: (1) a Bidirectional LSTM-Autoencoder with attention mechanisms to capture sequential feature dependencies; (2) a hybrid architecture combining deep learning embeddings (saved from this phase) with machine learning decision boundaries (Isolation Forest, OC-SVM); and (3) comprehensive cross-phase ablation studies comparing all models from Z-Score through hybrid architectures.

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